CryspLight: full simulation of monolithic crystals and their readout

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Abstract

The CRYSP concept proposes the use of large monolithic crystals read out by large SiPMs connected to custom electronics. Two candidates are being evaluated, BGO and CsI —or CsI(Tl)—. The crystals can be operated at room temperature or cooled in order to increase their light yield, but cooler temperatures imply a longer scintillation decay time. Cold operation also reduces the dark current, which improves performance but has an impact on the SPE calibration. The large SiPMs (6×6 mm^2 by default) allow to reduce the number of electronic channels typically by a factor four w.r.t. conventional pixelated detectors, in exchange for a larger capacitance which increases parasitic noise and complicates calibration.

Three readout systems have been studied: a waveform digitizer (CAEN DT2745), and two ASICs, namely the Citiroc 1A and the Radioroc2, both designed by Weeroc and read out on CAEN boards. The digitizer is needed in real measurements to establish the scintillation lifetime. The Citiroc 1A offers less flexibility than the Radioroc2, but it is widely available, unlike the newer model.

The room-temperature performance of BGO and CsI(Tl) is of interest and require the simulation of the two crystals with the three readouts. Cold operation of BGO at 250 K has the merit of increasing the light yield by a factor 1.7 in exchange for multiplying the lifetime by a factor 2.8. Cold operation of pure CsI at 77 K is also important. It multiplies by two the light yield of CsI(Tl) and offers a better lifetime (a single exponential of about 1 μs, while CsI(Tl) has a long-term exponential and afterglow which complicate its operation).

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1 The problem

The baseline concept for CRYSP contemplates large monolithic scintillating crystals (48×48 mm^2 in cross section and two radiation lengths thick) read out by a 8×8 array of large SiPMs (6×6 mm^2). The crystals are polished, wrapped in Teflon and coupled through optical grease to the sensors. The default SiPM is the S14160-6050HS from Hamamatsu.

A SiPM of 6×6 mm^2 has a capacitance of about 2 nF, which slows its pulse and raises the noise of the amplifier that reads it. The typical dark rate is several MHz. A good modeling of the sensor and the readout electronics is essential to understand the performance of the crystals.

CRYSP considers as potential materials BGO and CsI (or CsI(Tl)). BGO is denser and has a larger photoelectric cross section, but its light yield at room temperature is modest, about 8000 photons/MeV. This light yield can be increased by a factor 1.7 by cooling the crystal to 250 K.

The performance is studied along three axes. The first is the material of the crystal: BGO, dense and with a modest light yield, and CsI(Tl), which gives about seven times more light with a slower decay. The second is the readout. A CAEN DT2745 digitizer records the waveform of every channel and integrates the whole pulse; two Citiroc 1A chips of 32 channels each, and a single Radioroc2 of 64 channels, shape the signal of every channel and store one value of it per event. The third is the temperature. At room temperature the light yield of BGO is low and the dark counts are high; cooling raises the yield and lengthens the decay, and lowers the dark counts by a factor of 2 every 9 K. One cold configuration is studied for now, BGO at 250 K, read out by the digitizer and by the Radioroc2. The Citiroc is left out of it: its shaping, at most 87.5 ns, keeps only 12% of the light of a 0.84 µs decay against 33% of the BGO light at room temperature, so that with 1.7 times more light it would store 36% less than at room temperature (Section 7.1). Table 1 gives the three configurations of crystal and temperature.

The performance of the crystals is expressed in terms of the efficiency with which the scintillation photons are turned into photoelectrons, the number of photoelectrons per keV, the energy resolution and the spatial resolution. The simulation starts from an optical Monte Carlo of the light transport in the crystal and adds a model of the SiPM, with its gain, crosstalk, afterpulses, dark counts and pulse shape, and a model of each readout, so that its output is the charge per channel that each readout would record. The document describes these models, the quantities extracted from the simulated output, the tests that validate the code and the origin of every parameter.

The performance is expressed by four quantities. The light efficiency is the fraction of the scintillation photons that become photoelectrons in the SiPMs, and with the light yield it gives the absolute response, the number of photoelectrons per keV deposited. These two are properties of the crystal, of the optical coupling and of the SiPMs, and do not depend on the readout. What a readout records is a charge per channel, which contains more than these photoelectrons —the

	BGO, room temperature	CsI(Tl), room temperature	BGO, 250 K
crystal $x \times y \times z$ (mm)	$48 \times 48 \times 22.4$	$48 \times 48 \times 37.2$	$48 \times 48 \times 22.4$
thickness in X_0	2 ($X_0 = 11.2$ mm)	2 ($X_0 = 18.6$ mm)	2
temperature	about 295 K	about 295 K	250 K
light yield (ph/MeV)	8000[a]	54 000	13 500[a]
decay time	300 ns, plus a minor 60 ns component	730 ns (64%) and 3340 ns (36%)	0.84 µs [b]
emission peak	480 nm	550 nm	480 nm
refractive index	2.15	1.79	2.15
intrinsic resolution, FWHM at 511 keV	5%[c]	5%	5%[c]
source	^{22}Na, on axis, 30 mm from the face opposite the SiPMs		
readouts	digitizer, Citiroc 1A, Radioroc2	digitizer, Citiroc 1A, Radioroc2	digitizer, Radioroc2

Table 1: The three configurations of crystal and temperature. [a] The value at room temperature is the adopted reference; the value at 250 K follows from the temperature dependence of the light yield measured by Gironnet et al., scaled to that reference. Their curve, digitised by eye, gives 1.69 times the yield at 295 K; the text of their paper (half of the maximum at 250 K, a factor of about 3.5 between 6 K and room temperature) implies about 1.75, so that the yield at 250 K lies between 13 500 and 14 000 ph/MeV and the lower value is adopted. [b] From the three-level model of Gironnet et al. for the decay time; their measurement loses precision below 1 µs, and the value may be 20 to 30% long. [c] Less well known than for CsI(Tl) and taken equal to it (Section 4).

crosstalk and the afterpulses that each of them causes, and the dark counts— and, for the two ASICs, only the fraction of the scintillation pulse that their shaping keeps. The photoelectron budget of each readout therefore goes from the photons emitted to the charge recorded, and back from the calibrated charge to the number of photoelectrons, once the crosstalk, the afterpulses, the dark counts and the stored fraction are taken out; that number, and not the number of fired cells, is the absolute response measured by the readout. The energy resolution is the width of the 511 keV photopeak in the sum of the 64 channels, and of the 1275 keV one. The spatial resolution is the precision with which the position of the interaction is reconstructed from the 64 charges of the event.

To quantify these four quantities the simulation needs three models. The first is a realistic model of the propagation of the light inside the crystal: the emission of the scintillation photons, with their spectrum and their decay, their reflection on the wrapped and polished faces, their absorption, and their passage through the grease and the window of the SiPMs onto the active area of a channel, with the gaps between the channels. The second is a realistic model of the SiPM: its detection efficiency as a function of the wavelength and of the overvoltage, its gain, its crosstalk and afterpulses, its dark counts, and the pulse that an avalanche produces at the input of the electronics, which the capacitance of the device shapes. The third is a realistic model of each readout, from the input of the amplifier to the value stored per channel: shaping, noise, trigger, peak detection, conversion and saturation for the ASICs; sampling, noise, trigger, gate and baseline for the digitizer. This document describes these three models and only them.

The optical part of the simulation already existed: CryspLight, an optical Monte Carlo code in Julia designed to run on a GPU, transports the scintillation photons in a wrapped crystal read by an array of SiPMs. Its initial implementation assumed an idealised readout, in which the SiPM array covers the whole face without gaps and the light is monochromatic and detected with a single efficiency. The simulation described here replaces that ideal face by the actual array, so that a photon may fall between two SiPMs, and lets it reach the active surface through

the grease and the window of the device. Each photon is given a wavelength, because the efficiency with which a SiPM detects it depends on it. Each SiPM becomes a device that turns its photoelectrons into an electrical signal, with its gain, its crosstalk and its dark counts. And the electronics that measures that signal is simulated, so that the result is the value each readout records. How these extensions are added to the existing code is described in Section 12.

1.1 The three stages of the simulation

The simulation runs in three stages, each one feeding the next through a file.

1. **Optical simulation.**
   - It generates the gammas from the source, follows them to their energy deposits in the crystal, emits the scintillation photons and transports them on the GPU until they are absorbed or reach a SiPM.
   - It writes one file per run with, for every event, the energy and position of each interaction of the gamma in the crystal, the number of scintillation photons emitted and the number detected, and one record for every detected photoelectron: the channel that detected it, the time of detection and the wavelength of the photon.
2. **SiPM and electronics simulation.**
   - It reads the photoelectron records and applies the response of the SiPM, which turns every photoelectron into a current pulse of a given shape, multiplied by the gain and accompanied by crosstalk and afterpulsing, and adds the dark counts of the device.
   - It then applies the chain of the readout. For the two ASICs the pulses are amplified and shaped, compared with the discriminator thresholds, their maximum is stored by the peak detector until the hold, and the stored value is converted to ADC counts. For the digitizer the pulses are compared with the trigger threshold and integrated in the gate and the baseline window, with the noise of the sampled signal, which gives directly the charge that the analysis of the waveform would produce (Section 7.2).
   - It writes one file per run with, for every event and channel, the charge in ADC counts on each path of the readout and the discriminator flag, in the format of Section 8, together with the energy deposits and the photoelectron counts of the first stage.
3. **Analysis.**
   - It reads that file and extracts from it the performance: the gain of each channel from its own dark-count spectrum, its pedestal and noise, the photoelectron budget, the energy spectrum of the sum of the channels and its resolution, and the maps of the response over the array (Section 9).

Writing a file between the stages separates the slow part of the simulation from the fast one. The optical transport takes minutes on the GPU for a run of 10^5 events; the second and third stages take seconds to minutes. A change in the parameters of the SiPMs or of the electronics, or the choice of another readout, therefore requires rerunning only the last two stages, and the photoelectron records are reused.

2 Geometry

2.1 Crystal and wrapping

The crystal is a rectangular box, 48×48 mm^2 in cross section and 22.4 mm (BGO) or 37.2 mm (CsI(Tl)) thick, polished on its six faces. Five faces are wrapped by hand in several layers of

PTFE tape. The tape does not touch the crystal everywhere, so the model places a thin layer of air between the two. A photon that reaches a wrapped face is first reflected or transmitted at the crystal-to-air interface according to Fresnel's equations; only the transmitted fraction reaches the Teflon, which returns it diffusely with a reflectivity of 0.99, the value measured by Janecek and Moses for three or more layers of tape. The polish is described with the UNIFIED model of Geant4: the normal to the surface seen by each photon deviates from the geometric normal by a Gaussian angle of 1.3° rms, a value obtained for crystals of the same finish when the optical simulation was calibrated.

The air gap has a consequence that dominates the light collection. A photon whose direction is beyond the critical angle (27.7° in BGO, 34° in CsI(Tl)) on the face it meets is totally reflected, and since the faces of the box are parallel, a photon beyond the critical angle on all three pairs of faces stays trapped until the roughness of the surface, Rayleigh scattering in the bulk or absorption changes its direction or ends its life. The roughness is therefore not a detail: it sets how fast the trapped light leaks towards the readout face.

2.2 Coupling and window

The sixth face is coupled through optical grease ($n = 1.42$) to the SiPM tiles, whose photosensitive surface is covered by 0.15 mm of silicone resin ($n = 1.57$, from the datasheet). A photon leaving the crystal through this face crosses two interfaces, crystal to grease and grease to resin, and at each one it is reflected or transmitted according to Fresnel's equations. The first interface dominates (from 2.15 or 1.79 down to 1.42); the second reflects 0.3% at normal incidence. A reflected photon returns to the crystal and continues its life. The step from the resin into the silicon is not needed, since the detection efficiency quoted by Hamamatsu includes it. The detection efficiency is applied once the photon is inside the resin above an active area, as a function of its wavelength and of the overvoltage (Section 6), and independently of its angle of incidence. This independence is an assumption: the datasheet measures the efficiency at normal incidence from air, whereas our photons arrive from the grease at all angles up to the critical one. It is listed in Section 13.

2.3 The SiPM array

The array is made of four Hamamatsu S14161-6050HS-04 tiles, each with 4×4 channels of 6.0 mm active width on a 6.2 mm pitch. The tiles are butted, and since the active area stops 0.2 mm short of the package edge, the active areas of two channels across a tile boundary are 0.4 mm apart. With the origin at the centre of the matrix, the 64 centres are at ± 3.2, ± 9.4, ± 15.6 and ± 21.8 mm in each coordinate.

This geometry differs from an ideal array in three ways. The matrix is 50.0 mm wide, so the outer channels overhang the crystal by 1.0 mm and only 5.2 mm of their width faces it. Inside the 48×48 mm^2 footprint the active area is 93.4% of the total, the rest being the 0.2 mm gaps between channels, the 0.4 mm seam that crosses the centre in both directions and the overhang. And a photon transmitted into the resin above one of these gaps meets the surface of the package between the channels, whose reflectivity is not known; the model takes it diffuse with a reflectivity of 0.5, and it is listed in Section 13.

The array enters the simulation as a list of active rectangles, each with its channel number, so that the same code describes this array, an ideal one or any future layout. The channel numbering is part of the configuration of each run (Section 8).

3 Source and gamma transport

The ^{22}Na source is a point on the axis of the crystal, 30 mm from the centre of the face opposite the SiPMs. In each decay it emits a gamma of 1274.5 keV and, in 90.3% of the cases, a positron

that annihilates into two back-to-back gammas of 511 keV. The directions are isotropic. The simulation generates whole decays, draws the directions of the 1275 keV gamma and of the 511 keV pair, and transports the gammas whose line meets the front face. The face covers 13% of the sphere seen from the source, so the 1275 keV gamma enters in 13% of the decays; the 511 keV pair enters in 25% of the decays with a positron, twice as often, because either partner can be the one aimed at the face, and never both. Keeping the gammas of one decay together matters for the spectrum: when the 1275 keV gamma and a 511 keV gamma both deposit energy in the same event, their energies add, which produces the sum peak at 1786 keV, and the Compton continuum of the 1275 keV line lies under the 511 keV photopeak with its real weight. The gammas arrive at angles of up to 49° at the corners of the face, so the depth of the interactions differs from that of a beam at normal incidence.

Inside the crystal the gammas are followed by the transport code of PTCrysp, which simulates photoelectric absorption and Compton scattering with cross sections tabulated up to 10 MeV. Pair production, open above 1.022 MeV, is treated as a local absorption; at 1275 keV it is below 1% of the total cross section. Three effects are not simulated: the fluorescence that follows a photoelectric absorption on the K shell of bismuth, whose 77 keV X-ray can escape when the interaction is within a fraction of a millimetre of a surface and produces an escape peak at 434 keV; the Doppler broadening of the Compton scattering; and the scattering of the gammas in the source holder and in the walls of the cryostat. They are listed in Section 13.

4 Scintillation

Each energy deposit emits a number of photons drawn from a Poisson distribution whose mean is the light yield of the material times the energy deposited (Table 1). The crystal itself broadens the photopeak beyond the photon statistics. The light yield per unit energy of a scintillator depends on the energy of the electron that deposits it, and a gamma of a given energy is converted into electrons of different energies depending on how it interacts (one photoelectron, or a chain of Compton electrons followed by a photoelectron), so the total light fluctuates from event to event even for the same deposited energy. This intrinsic resolution is known for CsI(Tl), close to 5% FWHM at 511 keV, and less well for BGO. The non-proportionality curve of BGO has been measured (Gentile et al., 2015, and the references therein): relative to 662 keV, the light per keV is 73% at 23 keV, 89% at 60 keV, about 92% between 80 and 120 keV, 93% at 186 keV and 98.6% at 356 keV. A 511 keV deposit that proceeds through Compton electrons of 100 to 300 keV therefore yields a few percent less light than a single photoelectron, which is the origin of an intrinsic term of the same order as in CsI(Tl). The simulation takes 5% FWHM for both crystals, as a parameter, and applies it as a Gaussian smearing of the light of each event; the direct measurements that separate this term from the photon statistics (Moszynski et al.) remain to be checked. A more faithful treatment is possible, because the gamma transport records every deposit with its energy: weighting the light of each deposit by the curve at the energy of its electron produces the Compton-chain part of the intrinsic resolution for every line and for the continuum, leaving only the fluctuations within a single electron track to a smaller smearing. It is planned as a refinement.

Each photon is emitted at a time after the deposit drawn from the decay profile of the material: a single exponential for BGO, and for CsI(Tl) the sum of a 730 ns component carrying 64% of the light and a 3340 ns component carrying the rest. The two components of CsI(Tl) matter for the electronics, whose shaper is much shorter than either, because the stored signal follows the rate of arrival of the photons at the beginning of the pulse, which the fast component sets. For BGO at 250 K the decay time is the 0.84 µs of the model of Gironnet et al., in a material description of its own, so that the values of the calibration against the Ding cube stay with it.

Each photon is also given a wavelength drawn from the tabulated emission spectrum of the

material. The refractive index of the crystal, the Rayleigh scattering length (which scales as λ^{-4} from the calibrated value at the emission peak) and the detection efficiency of the SiPM are then evaluated at that wavelength. The bulk absorption length keeps its calibrated scalar value.

5 Optical transport

The photons are transported one by one on the GPU, each with its own stream of random numbers, so that a run is reproducible photon by photon whatever the number of threads. A photon moves in straight lines inside the box; at each step the code decides whether it is absorbed in the bulk or scattered by Rayleigh scattering before reaching a face, and at the face it applies the surface model described in Section 2: Fresnel reflection or transmission with the roughened normal, the air gap and the Teflon on the wrapped faces, the grease and the resin on the readout face. The photon ends its life when it is absorbed in the bulk, in the wrapping, in the package between the SiPMs or in a SiPM without producing a photoelectron, or when it is detected. The code checks that every photon ends in exactly one of these ways.

Three extensions of the transport are needed for the actual detector, and all three have been described above: the array of active rectangles with the reflecting surface between them, the second interface and the resin layer (the lateral displacement of the photon inside 0.15 mm of resin is negligible against the 6 mm pitch and is not tracked), and the wavelength of each photon with the tabulated refractive index, Rayleigh length and detection efficiency.

The optical simulation writes one file per run. For every event it records the energy and position of each interaction of the gammas in the crystal, with the number of interactions and the type of the first one, the number of scintillation photons emitted and the number detected, and the detected photoelectrons as a list of records of channel, detection time and wavelength, stored consecutively for all events with an index giving where each event starts. A record occupies 7 bytes, so a 511 keV event takes about 70 kB in CsI(Tl) and 15 kB in BGO, and a run of 10^5 events per configuration takes a few gigabytes of disk and a few minutes on the GPU.

6 Response of the SiPM

The SiPM is the S14160-6050HS element of Hamamatsu (datasheet KAPD1064E, November 2024); the S14161 tiles carry the same element. The model describes each channel separately.

Breakdown voltage and overvoltage. The breakdown voltage is 38.0 V at 25 °C and rises by 34 mV per kelvin, so it is near 36.4 V at 250 K and 34.5 V at 196 K; a measurement on one of our channels at that temperature gave 34.3 V. Within a tile the channels differ by 0.1 V typically and 0.2 V at most. The simulation draws the breakdown voltage of each channel from that spread, or takes it from the configuration of the run. The overvoltage of a channel is its bias minus its breakdown voltage, after the per-channel trim of the ASIC input DAC if it is used.

Gain. The charge of one avalanche is proportional to the overvoltage: 0.93×10^6 electrons per volt, which gives 2.5×10^6 at 2.7 V.

Detection efficiency. The probability that a photon arriving on the active area produces an avalanche is the product of a spectral factor, read from the datasheet curve and normalised to one at its maximum of 450 nm, and the efficiency at 450 nm as a function of the overvoltage: 37% at 1.7 V, 50% at 2.7 V, 54% at 3.2 V and constant above, because the probability of triggering an avalanche saturates.

Crosstalk and afterpulsing. An avalanche emits a few photons that can fire a neighbouring cell at the same time. The probability grows linearly with the overvoltage, from 2.5% at 1.6 V to 13% at 3.7 V. The model treats it as a branching process: every avalanche, primary or itself produced by crosstalk, fires a Poisson number of neighbours of mean $\lambda = -\ln(1-p)$, so that the probability of at least one extra avalanche is the datasheet value p and the total number of avalanches per primary follows a Borel distribution of mean $1/(1-\lambda)$ [7]. Compared with a geometric number of the same p, the branching gives more large groups: at 19.5% (5 V) the probability of seven or more extra avalanches is 3.5×10^{-4} against 1.1×10^{-5}, which matters for the rate of dark triggers at thresholds of several photoelectrons. Afterpulsing, the delayed release of a carrier trapped during the avalanche, is only qualified as low in the datasheet; measurements on devices of the same technology give about 1% at 3 V, growing with the square of the overvoltage, with delays of tens to hundreds of nanoseconds. The model uses 1% at 3 V with a time constant of 100 ns, as an adjustable parameter, and reduces the charge of the afterpulse by the fraction of the recharge of the cell completed at its delay, with the recovery time given with the pulse shape below. Its effect is a correlated addition of about one per cent to the charge of every channel.

Dark counts. Thermal carriers start avalanches at a rate that grows with the overvoltage like the detection efficiency and with the temperature by a factor 2 every 9 K. The datasheet dark current at 25 °C and 2.7 V, 2.5 μA, corresponds to 6×10^6 avalanches per second per device once the crosstalk is taken out, and the temperature law gives about 160 kHz per channel at 250 K and 5 V. The law is an educated guess, anchored on the datasheet value at room temperature only: at 196 K and 6.5 V it predicts about 2.5 kHz per channel, below the 8 to 15 kHz measured on this array, whose temperature reading is uncertain. A law anchored on that measurement instead would give 300 to 370 kHz per channel at 250 K rather than 160 kHz, and the threshold of the digitizer in the cold configuration would move from 320 to 384 counts to keep the rate of dark triggers below 1 kHz; the law is kept until the dark rate is measured against the temperature, and the sensitivity is listed in Section 13. The model adds dark avalanches, with their crosstalk and afterpulses, over the whole acquisition window of every event. The electronics hold their output at a fixed level, so the mean current of the dark avalanches gives no offset: the model subtracts the mean dark charge before the shaping and keeps only its fluctuations. Behind a charge-sensitive preamplifier this subtraction places the output of a channel with no avalanche below the pedestal of a random trigger, by the mean dark charge integrated over the response of the shaper: about 1.5 avalanches on the Radioroc2 at 100 ns of shaping and room temperature, 0.05 at 250 K, and nothing behind the voltage preamplifiers of the Citiroc, which pass no constant current. The calibration on the dark counts counts its photoelectron peaks from that level (Section 9). Every record is preceded by a warm-up, four peaking times of the longest shaper and ten fall times of the pulse, in which the dark avalanches and the noise already run through the filters, so that the response is stationary from the first sample of the record.

Pulse shape and cell occupancy. An avalanche discharges its cell, which recharges through its quenching resistor with the time constant $\tau_{\mathrm{rec}} = R_q C_d$ of the resistor and the cell capacitance. The recharge current flows into the input resistance R_{in} of the electronics, in parallel with the capacitance of the whole device, $C_t = 2000$ pF for the S14160-6050HS [1], which filters it with the time constant $R_{\mathrm{in}} C_t$ [4]. The current in the input is therefore the difference of two exponentials, rising with the shorter of the two time constants and falling with the longer one; the fast component that passes through the parasitic capacitance of the quenching resistor is neglected. The datasheet does not give τ_{rec}. A fall of about 60 ns has been reported for this device, measured through an amplifier with a pole-zero circuit that shortens the pulse [5], and the 6 mm devices of the series decay more slowly than the smaller ones because of their larger quenching resistance [6]. The model takes $\tau_{\mathrm{rec}} = 50$ ns, and $R_{\mathrm{in}} = 50\ \Omega$ for the three readouts:

it is the termination prescribed by the Citiroc datasheet [2], the DT2745 is assumed in its 50 Ω single-ended version, and for the Radioroc2 it is the external termination of one of its two input arrangements (Section 7.3); the other arrangement presents 100 Ω at the input of the chip and would give a slower pulse, peaking at 92 ns with a fall of 200 ns. With 50 Ω, $R_{\mathrm{in}}C_t = 100$ ns sets the fall, and the pulse peaks 69 ns after the avalanche and is 176 ns wide at half maximum. The input resistance is a setting of the run, so that once τ_{rec} is known the model computes the pulse at the input of each readout from its termination. A device has 14 331 cells, and a cell that has just fired needs τ_{rec} to recover; the model keeps track of this, although only the brightest CsI(Tl) channel comes within a percent of being affected.

The output of this stage, for every channel, is a train of avalanches with their times and charges, which is the input of the electronics.

7 Electronics

7.1 The Citiroc 1A

Two chips read the 64 channels, 32 each, with the channels assigned by the configuration of the run. The chain of each channel, as the datasheet (DS7004, version 2.5) describes it and as the simulation reproduces it, is the following.

1. **Input.** The train of avalanches becomes a voltage across the termination of the input, which the datasheet specifies as 50 Ω in series with 100 nF; with the 2 nF of the device it sets the fall of the pulse to 100 ns (Section 6). The input DAC of the channel does not touch the signal; it shifts the bias of the SiPM and therefore enters the model through the overvoltage.

2. **Preamplifiers.** Two voltage preamplifiers work in parallel on each channel: a high-gain one, with 15 pF of input capacitance and a gain adjustable between 10 and 600, and a low-gain one, with 1.5 pF and a gain between 1 and 60. The simulation gives each channel its own gain, drawn around the nominal value with an adjustable spread.

3. **Shapers.** Each preamplifier feeds a CR-RC^2 shaper with a peaking time between 12.5 and 87.5 ns, set in steps of 12.5 ns for the whole chip, separately for the high- and the low-gain shaper, each of which also has its own hold signal. A CR-RC^2 shaper is one differentiating stage followed by two integrating ones, all with the same time constant $\theta = t_p/2$, from 6.25 to 43.75 ns over the available settings. The preamplifiers are AC-coupled voltage amplifiers [2]: they amplify the voltage that the current of the SiPM develops across the 50 Ω termination, not its integral, so the differentiating stage acts on the current pulse itself. The impulse response of the shaper is then the derivative of the semi-Gaussian $(t/\theta)^2 e^{-t/\theta}$, and the shaped signal of an event is the convolution of the current of the avalanche train with it: a slow input is suppressed, and of the 100 ns pulse of our device the shaper keeps mostly the rising edge. Behind a charge-sensitive preamplifier, as on the Radioroc2, the differentiation cancels the integration of the preamplifier and the response to one avalanche is the semi-Gaussian itself; an earlier version of this model treated the Citiroc that way. The datasheet states the order of the shaper and nothing more; the exact response is a parameter to be checked against a single-photoelectron waveform if one can be probed on the test board.

4. **Noise.** A Gaussian noise series, shaped by the chain, is added after the shaper. With 2 nF of device capacitance at the input the series noise of the preamplifier dominates; it grows in proportion to the capacitance at the input and falls as the inverse square root of the peaking time [8]. The datasheets give one point each: a signal-to-noise ratio of 10 for one photoelectron at a gain of 10^6 for the Citiroc, with neither the capacitance nor the shaping

time stated, and a ratio above 10 on the high gain for a single photoelectron of 160 fC for the Radioroc2 [3], that is a noise below 16 fC, with no capacitance attached (the 100 pF of the same datasheet qualifies its timing jitter). If that noise grows in proportion to the capacitance at the input, 2 nF, about twenty times the 100 pF of a typical test, allows up to about 320 fC, or 0.7 photoelectron at the gain of 2.8×10^6 of 3 V; the bound does not fix the noise, it only leaves it open between a few hundredths and 0.7 photoelectron. The simulation assumes 0.1 photoelectron rms on the high-gain path at the longest shaping of the chip, and scales it as $(87.5\ \text{ns}/t_p)^{1/2}$ at shorter peaking times; the low-gain path has the same noise in ADC counts, ten times more in photoelectrons. The value is to be measured on the bench, as the pedestal and its width with the SiPMs biased below breakdown; the calibration of the chips on their dark counts at a short shaping depends on it (Section 13).

5. **Trigger.** A fast shaper with 15 ns peaking time, on the high-gain path, feeds two discriminators whose thresholds are common to the chip (10 bits) with a 4-bit trim per channel. One of them provides the trigger output of the channel; the other fills the hit register and the OR of the 32 channels. The simulation records, for each channel, every time the fast-shaped signal crosses the threshold, the discriminator re-arming when the signal falls below it; the trigger of the acquisition is then a configurable function of these times (the OR of one chip, the OR of both, or the earliest time at which N channels have crossed within a coincidence window), as the FPGA builds it in the real system.

6. **Peak detector and hold.** When the chip triggers, each channel starts storing the maximum of its shaped signal, on the high-gain and the low-gain path separately, and keeps it until the hold arrives, a fixed delay after the trigger (up to 500 ns). Each of the 64 channels stores its own maximum, and a channel that receives little light stores the maximum of its noise within the hold window; this gives a positive bias to the dim channels which the simulation reproduces because it stores the maxima the same way. At room temperature the maximum is set by the dark pulses and their crosstalk companions rather than by the electronic noise. The zero of the low-gain path, on which the energy is measured, is taken from self-triggered dark-count events as the mean low-gain value of the channels whose discriminator did not fire (Section 9), and subtracted channel by channel. It is not exactly an additive constant: in a channel that receives light the stored maximum is that of the signal, which a dark pulse changes only if it falls on top of it. The energy scale is therefore taken from the 511 and 1275 keV peaks, which absorbs the remaining offset. The alternative mode of the chip, in which the signal is sampled at a fixed time instead of at its maximum, is available in the simulation as well.

7. **Readout and conversion.** The stored values are read out through a multiplexer at up to 5 MHz and digitised by an external ADC. The simulation applies the gain and offset of the ADC, its quantisation (12 bits by default) and the saturation of the chain, which is reached either at 400 pC of input charge or at the full scale of the ADC, whichever comes first on each path. A high gain at full scale is not a saturation of the measurement, since the energy is read on the low gain; a channel is flagged when its low gain saturates. The dead time of the readout is recorded but does not reduce the number of events, because the rates of the simulated source are low.

The stored value of a channel is the peak of the shaped signal, calibrated in photoelectrons by the peak that one avalanche produces. For a scintillation pulse much longer than the peaking time the shaped signal does not reach the amplitude that all the photoelectrons would give if they arrived together: it is the convolution of the arrival rate with the pulse of one avalanche and with the shaper response, and its maximum, divided by the maximum of the same convolution for a single avalanche, is the fraction of the photoelectrons that the measurement keeps. With

the peaking time at 87.5 ns and the avalanche pulse of Section 6 (peak at 69 ns, 100 ns fall), the fraction is 0.33 for BGO at room temperature, whose 60 ns component with 10% of the light raises it from the 0.29 of a pure 300 ns decay, 0.10 for CsI(Tl), whose 730 ns component carries 64% of the light, and 0.12 for BGO at 250 K with its 0.84 µs decay; the light stored from cold BGO, $13\,500 \times 0.12$, is therefore 36% less than from BGO at room temperature, 8000×0.33. For a decay time τ much longer than the peaking time the fraction is the ratio of an effective window to τ. Behind the voltage preamplifier the window is 115 ns for the pulse of the model, against 280 ns behind a charge-sensitive one, because the differentiation suppresses the slow part of the train while the single avalanche, which serves as the unit, keeps its rising edge. The pulse at the Citiroc input is therefore the parameter that fixes these fractions, and it is listed in Section 13. The rise of the scintillation itself, below a nanosecond in BGO and some tens of nanoseconds in CsI(Tl), is short compared with the window and changes nothing. The photon statistics of the energy measurement is that of this fraction and not that of all the photoelectrons; the simulation computes it with the 64 channels, their noise and their dark pulses.

7.2 The DT2745 digitizer

The CAEN DT2745 samples its 64 channels every 8 ns with 16 bits over a full scale of 4 V, so one count is 61 µV, through an analog bandwidth of 20 MHz and a gain programmable between 1 and 100 for each group of 16 channels. Each SiPM is connected directly to the input of one channel, 50 Ω single-ended in the version assumed here (100 Ω differential in the other); there is no preamplifier and no shaper. The pulse of one avalanche is that of Section 6; the 20 MHz bandwidth of the digitizer, an 18 ns rise, is short compared with its rise and is neglected. Its area, the quantity the analysis measures, is 25 times its peak in count-samples. The height of the peak is a parameter, 3.2 counts per volt of overvoltage at gain 1, as an assumption: the pulse of the model, carrying the charge of one avalanche into 50 Ω, gives about 0.6 counts per volt, five times less (Section 13). Each channel fires when its sampled signal rises above its baseline by a threshold in counts, and the acquisition takes an event when N channels fire within a coincidence window; it then stores, for each of the 64 channels, a record of fixed length with a fixed number of samples before the trigger. The analysis of a record estimates the baseline of the channel as the mean of its pre-trigger samples and defines its charge as the sum of the samples in a gate that starts at the trigger and lasts long enough to contain the scintillation decay, minus the baseline times the number of samples in the gate. The single-photoelectron area, measured on dark pulses, converts that charge into photoelectrons.

The simulation of this readout does not build the waveforms. It computes, from the train of avalanches of each channel, the charge that the analysis would have extracted from the waveform, and writes it in the same event file as the Citiroc charges. This is possible because every operation between the avalanches and the charge is linear except the trigger. The sampled waveform of a channel is the sum of the pulses of its avalanches, each pulse scaled by the charge of its avalanche, plus the noise of the digitizer. The charge is a weighted sum of samples: weight one in the gate, weight minus the ratio of the gate length to the baseline length in the baseline window, zero elsewhere. Applying that weighted sum to the pulse of one avalanche gives a number that depends only on where the avalanche falls: its full area if it falls inside the gate, a fraction of it if it straddles an edge, minus the scaled area if it falls in the baseline window. The charge of the event is therefore the sum over the avalanches of the channel, scintillation and dark alike, of their charge times this number, plus the weighted sum of the noise samples. For white noise with an rms σ_s per sample the latter is a Gaussian with rms $\sigma_s\sqrt{N_g + N_g^2/N_b}$, with N_g and N_b the numbers of samples in the gate and in the baseline window, and one draw per channel and event is all the noise the charge needs. In units of photoelectrons this noise is 0.6 for a gate of 1 µs and 2 for a gate of 10 µs at 5 V of overvoltage and gain 1; the low-frequency noise and the correlation between the channels of one group observed with this digitizer are not in this model, and can be added as a second Gaussian common to the 16 channels of a group.

The trigger is the one non-linear step and the only one that needs the shape of the pulse. The simulation evaluates the sampled amplitude of each channel, the sum of the pulses of its avalanches on the 8 ns grid, around the times where avalanches occur, and records the first sample at which it exceeds the threshold. The trigger time of the event is then the time at which N of these crossings fall within the coincidence window, as in the acquisition, and it is the origin from which the gate and the baseline window are placed. The threshold decides which events are taken and, at room temperature, how often the dark pulses take one instead. The single dark photoelectron is not the problem: a dark avalanche fires neighbouring cells with the crosstalk probability, 19% at 5 V, so one dark pulse in 700 carries four photoelectrons or more, and over the 64 channels such pulses come about once per microsecond; a threshold of two or three photoelectrons triggers on them in every window. The emulation gives the efficiency for the photopeak and for the Compton events and the rate of dark triggers as a function of the threshold and of the majority; the threshold is the lowest whose dark-trigger rate stays below 1 kHz at 95% confidence. Its value depends on the pulse and on the crosstalk model and is given, for each readout, in the study at room temperature.

What goes into the event file is then the same as for the Citiroc: one charge per channel, as a 32-bit integer in count-samples, the set of channels that crossed the threshold in place of the hit register, and the trigger time. The single-photoelectron area and the noise per channel go into the calibration group. The saturation of the input, at a peak amplitude of 4 V divided by the gain, is checked on the brightest channel and flagged; it is not reached in the configurations studied. A subsample of events, a thousand per run, is also written as full waveforms, built with the same pulses and noise, for figures; they are not used by the analysis.

The gate is a choice of the analysis that the simulation informs, in its position as in its length. The emulation scanned both: opening the gate earlier than 50 ns before the trigger brings nothing, because the trigger fires within a few tens of nanoseconds of the first light, and every additional nanosecond of gate collects dark pulses; at 5 V of overvoltage the scan, made with an earlier and faster model of the pulse, gave a best gate of 1 µs for BGO at room temperature and 3 µs for CsI(Tl), and it is to be repeated with the present one. The study takes 5 V of overvoltage and gain 4 of the digitizer, where one photoelectron is 64 counts against a nominal noise of 4 counts per sample. The gain is the decisive choice for the calibration, not for the energy: at room temperature each channel produces 4.4 million dark counts per second at 5 V, 5.5 million fired cells with their crosstalk, one every 180 ns on average, and the simulated waveforms show that at gain 1, with 16 counts per photoelectron, the one- and two-photoelectron peaks of the dark-pulse spectrum are too shallow to fit channel by channel, whatever the estimator of the amplitude. At gain 4 a matched filter (the samples weighted with the pulse of one cell, after subtracting the mean of the four preceding samples) gives clean peaks, with a width of 6% of the gain and a fitted gain that follows the true one to 0.1% between channels. The length of the gate is the other choice. The gate must contain the scintillation decay, but at room temperature every channel produces dark counts at about 4.4 MHz, and in a gate of 10 µs the 64 channels gather about 2800 of them, 3500 fired cells with their crosstalk, against 1650 photoelectrons of signal in BGO: their fluctuation alone is 4% rms. A gate of 1 µs for BGO at room temperature, whose decay is 300 ns, and of 4 to 5 µs for CsI(Tl) brings that term below 1%. At 250 K the dark rate of the model is about 30 times lower and the gate can take the whole 0.84 µs decay with no penalty: the digitizer, with a gate of 4 µs, stores 0.99 of the photoelectrons of cold BGO, where the Citiroc would store 0.12 of them, against 0.33 at room temperature (Section 7.1), and the Radioroc2, at its longest shaping, 0.84. The threshold of the cold configuration, 320 counts, is fixed by the dark-trigger rate under the dark-rate law of Section 6; a rate anchored on the measurement at 196 K would raise it to 384 counts.

7.3 The Radioroc2

A single chip reads the 64 channels (datasheet version 2.0, January 2025). Its chain has the structure of the Citiroc one, with the differences listed below, and the simulation applies the same code with these differences as settings.

1. **Input.** The SiPM is connected to the input of the chip through one of two input DACs, which also set its bias channel by channel. One of them presents 100 Ω at the input of the chip, so that the SiPM is connected without an external resistor; the other has a high output impedance and requires an external resistor in series with 100 nF to ground. The simulation takes 50 Ω for now, an external termination of the second kind, as for the Citiroc, until the board is chosen. The 100 Ω arrangement would move the peak of the pulse of one avalanche from 69 to 92 ns and its fall from 100 to 200 ns; the hold delay of the calibration at 100 ns of shaping would move from 229 to 268 ns, and the fraction of a 0.84 µs decay stored at 1.8 µs would hardly change, from 0.843 to 0.846. The input resistance is a setting of the run, so that the two arrangements can be compared.

2. **Preamplifiers and shapers.** Two charge preamplifiers per channel, with gains between 5 and 80 on the high-gain path and 0.5 and 8 on the low-gain one, each followed by a shaper whose peaking time is set channel by channel: 20 to 300 ns in steps of 20 ns, or, in the slow mode, 120 ns to 1.8 µs in steps of 120 ns. The order of the shaper is not stated; the simulation takes CR-RC^2, as for the Citiroc, as an assumption. The slow mode is the natural choice for CsI(Tl) and for cold BGO, whose 0.84 µs decay through 1.8 µs of shaping stores 0.84 of the photoelectrons (Section 7.1); at room temperature a long shaping also collects more dark pulses, about 14 photoelectrons per channel in the effective window of the longest setting, so that the shaping time has an optimum per crystal which the emulation gives.

3. **Noise.** As for the Citiroc: the datasheet bound, a noise below 16 fC on the high gain for a single photoelectron of 160 fC, with neither the capacitance nor the shaping time attached, allows up to about 0.7 photoelectron once scaled to the 2 nF of our device. The simulation assumes 0.1 photoelectron rms at 100 ns, scaled as $(100\ \text{ns}/t_p)^{1/2}$, until the bench measures it; the dark pulses, which the model generates explicitly, grow instead with the peaking time.

4. **Triggers.** Each channel has a time preamplifier of 200 MHz bandwidth with two discriminators, T1 and T2, of independent thresholds common to the chip with a per-channel trim, and a charge discriminator TQ on the output of the low-gain shaper. The time preamplifier sees the avalanche almost unshaped, so T1 and T2 respond to the instantaneous amplitude within a few tens of nanoseconds: for BGO at room temperature the brightest channel of a 511 keV event reaches only 4 to 5 photoelectrons there, while the dark pulses with crosstalk reach the same once per microsecond over the array, and a single time threshold cannot separate them. The charge trigger sees the integrated light of the event, hundreds of photoelectron-equivalents for the photopeak against a handful for a dark pulse, and is the trigger the simulation takes by default for the slow scintillators; T1 and T2, single channel or in a majority built by the FPGA of the board, are emulated as an alternative. The pulse at the time preamplifier is not the one measured through the 20 MHz digitizer; the simulation takes a 5 ns rise and a 20 ns fall as parameters, which only enter the T1 and T2 numbers.

5. **Acquisition trigger, hold and peak detectors.** The peak detectors of all channels are opened by one acquisition trigger, selectable among the chip-wide OR of T1, of T2 or of TQ, the local trigger of each channel, or an external signal; the hold follows it after a delay set on the chip between 2 ns and 2.6 µs, or comes from outside. Each path stores

the maximum of its shaped signal from the trigger to the hold, as in the Citiroc, with the same positive bias of the dim channels.

6. **Readout and conversion.** The stored values of the two paths leave through two analog multiplexers at 200 ns per channel, 12.8 µs for the 64 channels, to the ADC of the acquisition board, which the simulation takes as the 12-bit converter of the evaluation board. The low-gain path covers 2000 photoelectrons at a gain of 10^6, about 320 pC, at its default preamplifier gain of 0.53; the high-gain path, whose default gain is 5.3, saturates ten times lower. The ceilings scale inversely with the preamplifier gains, but the high gain cannot be set below 5, so its ceiling cannot exceed about 76 avalanches at our SiPM gain of 2.8×10^6, which the brightest channel of a 511 keV event in BGO at room temperature exceeds in a quarter to a half of the events; the chip is therefore operated as designed, both paths read and combined, the high gain for every channel below its ceiling and the low gain above it, and the simulation stores that dual-gain value. Each channel can instead route its shaped signal to a pin of its own, for a parallel digitisation that the simulation does not use.

8 Files

8.1 The photoelectron file

The optical simulation writes its photoelectron records in one HDF5 file per run. The records of all events are stored consecutively in three arrays, `channel` (one byte), `t_ns` (a 32-bit float) and `lambda_nm` (16 bits), and an array `offset` of length one more than the number of events gives the position of the first record of each event. The event index is the same as in the file of interactions and photon counts written by the same run. The attributes of the file record the configuration of the run and the version of the code.

8.2 The event file

The electronics simulation writes one HDF5 file per run with the structure of Table 2. Arrays are stored event-major, so that a reader obtains an array of n events by 64 channels without transposition. The structure holds only what a readout records, apart from the group `/mc`, so that the analysis of the performance uses the same information a real system would have.

9 Observables

The analysis extracts the performance from the event file of each configuration and readout, using only what the readout records, and compares it with the quantities of the group `/mc`.

1. **Single-photoelectron spectra.** Built from dark pulses, channel by channel, each readout with its own data only, they give the gain of each channel and its spread, as a system with many crystals would have to calibrate itself with its own readout. For the two chips every crossing of the discriminator of a channel at half a photoelectron holds the high gain of that channel at the maximum of its pulse, the hold delay being the one that maximises the held value; the shaping is short enough for the dark pulses not to pile up under the held value. The 1, 2 and 3 photoelectron peaks of the spectrum, the higher ones from crosstalk, give the gain as the slope of the straight line through their positions and the zero of the high gain as its intercept. The number of photoelectrons of each peak is counted from the level at which a channel with no avalanche is held, which is not the pedestal of a random trigger: the chip removes the mean current of the dark counts, so that level lies below the pedestal by the mean dark charge integrated over the response of the shaper (Section 6), 1.5 photoelectrons on the Radioroc2 at 100 ns of shaping and room temperature, where

group	content
`/run`	Attributes describing the run: crystal and material, temperature, bias and breakdown voltage per channel, overvoltage per channel; for the ASICs the discriminator thresholds, hold delay, shaping times, preamplifier gains, ADC and trigger definition; for the digitizer its gain, record length, pre-trigger, threshold and majority, and the gate and baseline window used to define the charge; the full configuration of the simulation with the origin of every parameter, and the version of the code.
`/channels`	One entry per channel: the position of its centre in millimetres, its tile, its chip and channel number on the chip, and its status (live, dead or noisy). This group ties the readout channel to the geometry of Section 2.3.
`/calib`	The single-photoelectron gain of each channel, in ADC counts per photoelectron, and the mean and rms of its pedestal, on each recorded path, as used by the simulation; the calibration of each readout on its own dark counts (Section 9) writes its fitted values beside them.
`/events`	For every event: the charge of the 64 channels, as 16-bit integers of ADC counts on each path for the ASICs and 32-bit integers of count-samples for the digitizer; the hit register of the 64 channels, which for the digitizer is the set of channels that crossed the threshold; the trigger time; and flags for saturation and for pile-up.
`/mc`	The quantities written by the optical simulation for the same events (positions and energies of the interactions, photons emitted and detected, the integrated photoelectron map), so that any observable can be examined against them.

Table 2: Structure of the event file.

the one-photoelectron peak therefore lies below the pedestal. The peaks are numbered from that expected zero, computed from the dark rate and the shaper, and a spectrum whose peaks are not equally spaced at the expected positions gives no gain rather than a wrong one. The zero of the low-gain path is the mean low-gain value of the channels whose discriminator did not fire, in self-triggered dark-count events taken with the settings of the run, and its gain is the high-gain one times the ratio of the two paths. For the digitizer a matched filter on the waveform gives the one-photoelectron peak, the baseline being subtracted; the estimator takes the local maxima of the filtered waveform over all alignments of the template, which for the long pulse of the device exceed the amplitude of an isolated pulse by 4 to 6% depending on where the avalanche falls between two samples, so it is normalised to give one on average over the sampling phases for an isolated single cell. The fitted gains are compared channel by channel with the gains the simulation used.

2. **The two paths of the chips.** On both chips the high-gain path gives the single-photoelectron peaks and the low-gain path the energy. The peaks are usable only at a short shaping of the high gain, where the dark pulses do not pile up under the held value. With the pulse and noise of the model, the Radioroc2 calibrates every channel at 50 and 100 ns (the fitted gains scatter by 3.6% between channels at 100 ns, the setting adopted) but not at 300 ns or at 25 ns, where the noise hides the peaks; the Citiroc calibrates every channel at 25, 50 and 87.5 ns, with a scatter of 2.2, 5.1 and 7.5%, and its high gain is set to 25 ns while the low gain stays at 87.5 ns for the energy. The gain of the low-gain path follows from the ratio of the two paths at their two peaking times, which the charge-injection input of each chip measures (the Citiroc has one on each preamplifier, the Radioroc2 a 1.5 pF capacitor per channel). The simulation assumes this ratio known. The injected charge arrives at once, while the charge of an avalanche is spread over the 100 ns of its pulse, so that at a short shaping the high gain responds less to an avalanche than to the same injected charge; the ratio measured by injection has to be corrected for this

with the pulse shape. The 1275 to 511 ratio of the low-gain sum then tests the linearity of that path alone.

3. **Pedestals and noise.** The rms of the pedestal of each channel and of the sum, which enter the energy resolution.

4. **The photoelectron budget.** For the photopeak at 511 keV: the scintillation photons emitted; the fraction that reaches the active area of a channel and the fraction of those detected, whose product is the light efficiency; the photoelectrons per keV, which is the absolute response. Each readout then records, in units of the single-photoelectron gain, a charge that contains the crosstalk and the afterpulses of these photoelectrons and the dark counts in its window, and that the two ASICs reduce to the fraction of the pulse their shaping keeps. The budget goes back from this calibrated charge to the photoelectrons: the mean of the dark counts is subtracted, the charge is divided by the stored fraction, and the result by the mean number of fired cells per photoelectron, which for the branching crosstalk of Section 6 is the Borel mean $1/(1-\lambda)$ with $\lambda = -\ln(1-p)$ (1.28 at the 19.5% of 5 V, where $1/(1-p)$ would give 1.24), times the factor of the afterpulses. The number obtained is the absolute response measured by the readout, and it is compared with the number of photoelectrons in `/mc`. The ratio of the photoelectrons between the BGO configurations at room temperature and at 250 K gives the change of the light yield with the temperature.

5. **The energy spectrum** of the equalised sum of the 64 channels: the 511 and 1275 keV lines, the sum peak, the Compton continua and the backscatter region. The ratio of the two photopeak positions, 2.5 if the chain is linear, tests the dynamic range and the trigger; for the peak-detector chips the energy scale is the straight line through the two peaks, since the pedestal of the peak detectors leaves an offset. The width of the 511 keV line is decomposed into the noise of the pedestals, the statistics of the stored photoelectrons, the fluctuation of the crosstalk, the afterpulses and the dark counts, and the intrinsic resolution put into the simulation.

6. **The spatial resolution.** The position of the interaction is reconstructed from the 64 charges of the event, with the barycentre of the light and with the neural network developed for the idealised readout, and compared with the position in `/mc`. The reconstruction on the output of each readout is not done yet (Section 13).

7. **The brightest channel.** The distribution of the largest charge in an event, and its ratio to the total, describe the light distribution near the readout face; in CsI(Tl) its tail shows the saturation of the chain.

8. **Channel maps.** The fraction of events in which each channel is hit and the mean charge it collects show the response of the outer channels (which see only 5.2 mm of crystal), of the central seam and of the tile boundaries. The reflectivity of the surface between the channels and the roughness of the crystal are the two parameters that change these maps.

9. **The seed-channel map.** The position of the 511 keV photopeak for the events grouped by their brightest channel, in units of the median over the channels: a map of the non-uniformity of the light collection.

10. **Dependence on the overvoltage**: light, crosstalk, resolution and saturated fraction against the overvoltage, with the curves of Section 6.

11. **The readouts on the same crystal.** The photoelectrons at 511 keV recorded by the digitizer, which integrates the whole decay, and by the Citiroc or the Radioroc2, which store a fraction of them, are in the ratio computed in Section 7.1 from the pulse shape

and the shaping time: 0.33 for BGO and 0.10 for CsI(Tl) with the Citiroc, 0.97 for BGO and 0.71 for CsI(Tl) with the Radioroc2 at 1.8 µs, at room temperature. The ratio follows from the shaping chains alone, with no free parameter.

12. **Trigger efficiency and gate length.** The photopeak content and the rate of dark triggers against the threshold, for each readout, and for the digitizer the resolution against the length of the gate; these curves fix the settings of each readout.

10 Validation

The code is tested at each of its extensions, with tests that have an analytic answer or reduce a new feature to an old one.

- The array described as active rectangles reproduces, with the rectangles of an ideal array, the result of the ideal array; with the gaps and no reflection on the surface between the channels, the fraction of photons lost under uniform illumination equals the geometric dead fraction.
- When the grease and the resin have the same index the second interface reflects nothing; when the resin has the index of the crystal and there is no grease, the result is that of a crystal in optical contact with the sensor.
- A spectrum reduced to a single wavelength reproduces the monochromatic simulation.
- The photoelectron records, binned in time, reproduce the time-binned maps written by the run.
- The rate of gammas on the front face equals the solid-angle fraction, and the distribution of their incidence angles the analytic one.
- A single avalanche through the shaper gives the analytic CR-RC^2 peak at t_p; a Poisson train with a long decay time gives the stored fraction of Section 7 within its statistics; a peak detector fed with noise only gives the expected bias of the maximum of a Gaussian series; the saturation clips at the configured charge.
- The pulse of one avalanche peaks at $\tau_a \tau_b \ln(\tau_a/\tau_b)/(\tau_a - \tau_b)$ and falls with the longer constant; the branching crosstalk gives no extra avalanche with probability $1 - p$ and the Borel probabilities for the larger groups; the average of the shaped dark counts is zero once their mean charge is subtracted, from the first sample of the record; a single avalanche held at its maximum after its own crossing gives the pedestal plus one gain; a chain behind a voltage preamplifier passes no constant input.
- Isolated pulses of one, two and three cells embedded in stationary dark histories, at room temperature and at 250 K, receive their number of cells from the calibration on the dark counts, and the fitted gain is the true one within a few percent; the estimator of the digitizer calibration gives one on average over the sampling phases for an isolated single cell.
- The energy resolution of a simulated configuration agrees with the quadrature of its stated terms.

11 Parameters and their provenance

12 Implementation

The optical simulation is CryspLight, a Julia package whose transport kernel runs on the GPU of Apple computers through Metal and, identically, on the CPU. Its reference version of the transport and its GPU version are kept equal draw by draw, and a test proves that they produce the same result bit for bit on the CPU; every extension described here is written in both.

The state of the code before this work is kept under the git tag `pedsim`, and the extensions grow on the branch `fullsim`. Nothing in the existing behaviour is changed: each extension is switched on by a block in the configuration file of the run and switched off by its absence, in which case the code follows the old path with the same random numbers. A test runs a small configuration of the old kind after every change and compares its output with stored checksums. The existing configurations, output files and analysis scripts are left as they are; the new simulation has its own configurations under `fullsim_runs/`, grouped by study, its own analysis under `fullsim_analysis/` and its documents under `docs_fullsim/`. The analysis is written in Julia, a library of readers, fits, spectra and maps and one script per study; the scripts compute and write files, HDF5 for arrays and CSV for tables, under `fullsim_analysis/results/`, and Python scripts draw the figures from those files, so that a figure is remade without recomputing and every number cited in a document is in a file produced by a tracked script. The SiPM and electronics simulation is a new module, `src/frontend/`. Simulation outputs go to the directory `output/`, which is not under version control.

The optical stage records every photon that reaches the resin above an active area, with the detection efficiency set to one in the transport, and the SiPM stage applies the efficiency by keeping each record with the probability of Section 6 for its wavelength and the overvoltage of the run. Thinning a Poisson process this way is exactly equivalent to applying the efficiency photon by photon during the transport, and it lets one optical run serve every overvoltage of a scan. The photoelectron file is written in chunks of decays as the run proceeds, so that a run of any length streams to disk; the random streams of the decays and of their photons are disjoint between chunks, and the gamma-level content of a run does not depend on how it was chunked.

The GPU imposes a few constraints on the extensions: the tables of the spectra and of the wavelength-dependent quantities travel in one buffer of 32-bit floats, the switches and scalars in one structure, and no new random number is drawn unless the corresponding extension is on.

13 Open items

- Whether the detection efficiency depends on the angle of incidence when the photons arrive from the grease rather than from air.
- The reflectivity and the specularity of the surface of the package between the channels.
- The light yield and decay time of BGO at 250 K, taken from Gironnet et al.: the decay time may be 20 to 30% long, the ratio of the yields at 250 and 295 K is 1.69 from the digitised curve and about 1.75 from the text of the paper (13 500 to 14 000 ph/MeV), and the yield changes by about 1% per kelvin near 250 K, so that the temperature of the crystal must be known to a few kelvin.
- The dark rate against the temperature. The factor of 2 every 9 K is an educated guess anchored on the datasheet value at room temperature; it predicts 2.5 kHz per channel at 196 K and 6.5 V where 8 to 15 kHz were measured on this array, with an uncertain temperature reading. A law anchored on that measurement gives 300 to 370 kHz per channel at 250 K instead of 160 kHz, and the cold digitizer threshold would move from 320 to 384 counts. The rate is to be measured against the temperature.

- The intrinsic resolution of BGO.
- For the Citiroc: the exact response of the shaper, the noise spectrum, the droop of the peak detector during the readout, and the pulse shape at the input of the preamplifier with a device of 2 nF.
- The noise of both chips, assumed to be 0.1 photoelectron rms on the high gain at the longest shaping where the datasheet bound allows up to 0.7. The calibration on the dark counts at a short shaping depends on it: with 0.7 photoelectron at 100 ns the Radioroc2 spectrum at room temperature still shows peaks, but they are not the photoelectron peaks, and a straight line through them returns a gain of 5.9 counts per cell for a true gain of 10. The shaping of the calibration is to be chosen again once the noise is measured.
- The calibration of the chips is simulated with every channel held after its own crossing. On a board with a common trigger and a common hold, the selection of pulses differs, and a test of that selection on the Citiroc moved the fitted gain by 3.2%; the local-channel calibration is an idealisation until the acquisition of the board is modelled, which waits for the bench measurements.
- The recovery time of the cells and the input resistance of each readout, which fix the pulse, and the peak height per photoelectron of the digitizer.
- For the Radioroc2: the order of the shaper, the noise, which of its two input arrangements the board uses (100 Ω at the input of the chip or an external termination, which change the pulse and the hold delay of the calibration), and the acquisition board and ADC to be used. Its dark-count calibration spectrum at room temperature shows a population below the pedestal of the random trigger: it is the one-photoelectron peak, held 1.5 photoelectrons below the pedestal because the mean current of the dark counts is removed before the shaping (Section 6); at 250 K the removed level is 0.05 photoelectron and the peak sits where expected.
- For the digitizer: the noise below 1 MHz and its correlation within a group of 16 channels, and whether the gain stage lowers the noise per photoelectron, which decides which gain serves the cold configuration, where the overvoltage may be lower.
- In the gamma transport: fluorescence and escape of the K-shell X-rays, Doppler broadening, and scattering in the source holder and in the cryostat.
- The temperature of the SiPMs relative to the crystal in the cold configuration, which sets their breakdown voltage and dark rate.
- The reconstruction of the position on the output of each readout, done so far only on the idealised readout.
- The photoelectron budget of Section 9, with the crosstalk, the afterpulses and the dark counts taken out, for every configuration and readout.

References

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parameter	value	source	status
crystal sizes	Table 1	setup	fixed
polish σ_α	1.3°	calibration of the optical simulation	fixed
Teflon	Lambertian, $R = 0.99$, air gap	Janecek & Moses	fixed
grease index	1.42	setup	fixed
resin index, thickness	1.57, 0.15 mm	datasheet	fixed
SiPM array	centres at $\pm 3.2, \pm 9.4, \pm 15.6, \pm 21.8$ mm	setup	fixed
reflectivity between channels	0.5, diffuse	assumption	parameter
BGO yield, room temperature	8000 ph/MeV	adopted reference; Gironnet curve, Moszynski	fixed
BGO yield 250 K	13 500 ph/MeV	Gironnet curve scaled to the room-temperature value: 1.69 times from the digitised curve, about 1.75 from the text of the paper, 13 500 to 14 000	about 1% per kelvin; digitisation
BGO decay 250 K	0.84 µs	Gironnet three-level model	possibly 20 to 30% long
BGO decay, room temperature	300 ns	literature	fixed
CsI(Tl) yield, decay	54 000; 730/3340 ns, 64/36%	calibration; literature	fixed
intrinsic resolution	5% FWHM at 511 keV, both crystals, scaling as $(511\ \text{keV}/E)^{1/2}$	CsI(Tl) measurements; BGO curve from Gentile et al.; Moszynski et al. to check; the energy dependence is an assumption	assumption for BGO and for the exponent
emission spectra	tabulated	literature	to digitise
breakdown voltage	$38\ \text{V} + 34\ \text{mV/K}\,(T - 298\ \text{K})$	datasheet; measurement at 196 K	fixed
gain, PDE, crosstalk vs overvoltage	Section 6; crosstalk as a branching (Borel) process	datasheet curves [1]; [7]	fixed
dark rate	6 MHz at 25 °C, 2.7 V; factor 2 every 9 K (2.5 kHz at 196 K and 6.5 V, against 8 to 15 kHz measured; 160 kHz at 250 K and 5 V, 300 to 370 kHz if anchored on the measurement)	datasheet; the temperature law an educated guess	educated guess, to measure against the temperature
afterpulsing	1% at 3 V, $\propto V_{\text{ov}}^2$, 100 ns	literature on the same technology	parameter
pulse shape	$\tau_{\text{rec}} = 50$ ns filtered by $R_{\text{in}}C_t = 50\ \Omega \times 2000$ pF: peak at 69 ns, 100 ns fall (100 Ω input of the Radioroc2: 92 ns, 200 ns)	C_t datasheet [1]; τ_{rec} assumption [5, 6]; R_{in} Citiroc datasheet, external termination assumed for the Radioroc2, 50 Ω version of the digitizer	τ_{rec} assumption; R_{in} setting of the run
digitizer scale	3.2 counts per photoelectron per volt of overvoltage at gain 1 (0.6 from the pulse model); area 25 times the peak	assumption; area from the pulse	uncertain by a factor of 5
digitizer noise	4 counts rms per sample, white	datasheet; measurement with this digitizer	fixed; low-frequency and group terms not included
digitizer settings	gain, record, pre-trigger, thresh-	study at room temper-	setting of the